Moments and Singular Perturbation

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1 Problem Setup

SYSTEM:

$$\dot{x} = A_{11}x + A_{12}z + B_1u \tag{1}$$

$$\varepsilon\dot{z} = A_{21}x + A_{22}z + B_2u \tag{2}$$

$$y = C_1x \tag{3}$$

No z term as otherwise we would have feedforward in the reduced model.
In state space form:

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B_1 \\ \frac{1}{\varepsilon}B_2 \end{bmatrix} u \tag{4}$$

SIGNAL:

$$\dot{w}_1 = S_{11}w_1 + S_{12}w_2 \tag{5}$$

$$\varepsilon\dot{w}_2 = S_{21}w_1 + S_{22}w_2 \tag{6}$$

REDUCED MODEL SIGNAL:

$$S_{22}w_2 + S_{21}w_1 = 0 \tag{7}$$

$$w_2 = -S_{22}^{-1}S_{21}w_1 \tag{8}$$

$$\dot{w}_1 = (S_{11} - S_{12}S_{22}^{-1}S_{21})w_1 \tag{9}$$

$$u = (L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \tag{10}$$

REDUCED MODEL OF THE SYSTEM:

$$A_{22}z + A_{21}x + B_2u = 0 \tag{11}$$

$$z = -A_{22}^{-1}\left(A_{21}x + B_2u\right) \tag{12}$$

$$\dot{x} = (A_{11} - A_{12}A_{22}^{-1}A_{21})x + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21})w_1 \tag{13}$$

FORWARD MOMENTS

$$M_g := \begin{bmatrix} C_1\Pi_{x1} & C_1\Pi_{x2} \end{bmatrix} \tag{14}$$

FORWARD SYLVESTER EQUATION FOR THE FULL SYSTEM:

$$\begin{bmatrix} A_{11} & A_{12} \\ \frac{1}{\varepsilon}A_{21} & \frac{1}{\varepsilon}A_{22} \end{bmatrix} \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} + \begin{bmatrix} B_1L_{g1} & B_1L_{g2} \\ \frac{1}{\varepsilon}B_2L_{g1} & \frac{1}{\varepsilon}B_2L_{g2} \end{bmatrix} = \begin{bmatrix} \Pi_{x1} & \Pi_{x2} \\ \Pi_{z1} & \Pi_{z2} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ \frac{1}{\varepsilon}S_{21} & \frac{1}{\varepsilon}S_{22} \end{bmatrix} \tag{15}$$

FORWARD SYLVESTER EQUATION FOR THE REDUCED MODEL

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})\tilde{\Pi} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) = \tilde{\Pi}(S_{11} - S_{12}S_{22}^{-1}S_{21}) \tag{16}$$

Can we show that as $\varepsilon \to 0$, the forward syvlester equation for the full system tends towards the forward sylvester equation for the reduced model?

2 Proof (I guess?)

Expanding out (14) we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1 L_{g1} = \Pi_{x1} S_{11} + \frac{1}{\varepsilon}\Pi_{x2} S_{21}, \tag{17a}$$
$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1 L_{g2} = \Pi_{x1} S_{12} + \frac{1}{\varepsilon}\Pi_{x2} S_{22}, \tag{17b}$$
$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2 L_{g1} = \Pi_{z1} S_{11} + \frac{1}{\varepsilon}\Pi_{z2} S_{21}, \tag{17c}$$
$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2 L_{g2} = \Pi_{z1} S_{12} + \frac{1}{\varepsilon}\Pi_{z2} S_{22}. \tag{17d}$$

Multiplying (16c) and (16d) by ε we have:

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1 L_{g1} = \Pi_{x1} S_{11} + \frac{1}{\varepsilon}\Pi_{x2} S_{21}, \tag{18a}$$
$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1 L_{g2} = \Pi_{x1} S_{12} + \frac{1}{\varepsilon}\Pi_{x2} S_{22}, \tag{18b}$$
$$A_{21}\Pi_{x1} + A_{22}\Pi_{z1} + B_2 L_{g1} = \varepsilon\Pi_{z1} S_{11} + \Pi_{z2} S_{21}, \tag{18c}$$
$$A_{21}\Pi_{x2} + A_{22}\Pi_{z2} + B_2 L_{g2} = \varepsilon\Pi_{z1} S_{12} + \Pi_{z2} S_{22}. \tag{18d}$$

We now consider the ε series expansion of Π_{x1}, Π_{x2}, Π_{z1}, Π_{z2}. We can see that by multiplying (17a) and (17b) by ε, and setting ε to 0 we have that the zeroth order term for Π_{x2} is 0.

We then write:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1 L_{g1} = \Pi_{x1}^{(0)} S_{11} + \Pi_{x2}^{(1)} S_{21} + O(\varepsilon), \tag{19a}$$
$$A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2} = \Pi_{x1}^{(0)} S_{12} + \Pi_{x2}^{(1)} S_{22} + O(\varepsilon), \tag{19b}$$
$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2 L_{g1} = \Pi_{z2}^{(0)} S_{21} + O(\varepsilon), \tag{19c}$$
$$A_{22}\Pi_{z2}^{(0)} + B_2 L_{g2} = \Pi_{z2}^{(0)} S_{22} + O(\varepsilon) \tag{19d}$$

Each of the 4 equations in (18) are essentially polynomials in epsilon.
After equating coefficients, we can write $\Pi_{x2}^{(1)}$ as such (Assuming S_{22} is invertible):

$$\Pi_{x2}^{(1)} = \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2} - \Pi_{x1}^{(0)} S_{12}\right) S_{22}^{-1} \tag{20}$$

Substituting (19) into (18a) we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1 L_{g1} = \Pi_{x1}^{(0)} S_{11} + \left(A_{12}\Pi_{z2}^{(1)} + B_1 L_{g2} - \Pi_{x1}^{(0)} S_{12}\right) S_{22}^{-1} S_{21} + O(\varepsilon) \tag{21}$$

Rearranging we have:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1 L_{g1} = \Pi_{x1}^{(0)}\left(S_{11} - S_{12} S_{22}^{-1} S_{21}\right) + \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2}\right) S_{22}^{-1} S_{21} + O(\varepsilon) \tag{22}$$

Similarly, from (18c) we can substitute in an expression for $\Pi_{z1}^{(0)}$:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}A_{22}^{-1}\left(\Pi_{z2}^{(0)} S_{21} - A_{21}\Pi_{x1}^{(0)} - B_2 L_{g1}\right) + B_1 L_{g1} = \Pi_{x1}^{(0)}\left(S_{11} - S_{12} S_{22}^{-1} S_{21}\right) + \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2}\right) S_{22}^{-1} S_{21} + O(\varepsilon) \tag{23}$$

Rearranging we have:

$$\begin{aligned} &\left(A_{11} - A_{12}A_{22}^{-1}A_{21}\right)\Pi_{x1}^{(0)} + \left(B_1 - A_{12}A_{22}^{-1}B_2\right) L_{g1} \\ &\quad + \left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - \left(A_{12}\Pi_{z2}^{(0)} + B_1 L_{g2}\right) S_{22}^{-1}\right) S_{21} \\ &= \Pi_{x1}^{(0)}\left(S_{11} - S_{12} S_{22}^{-1} S_{21}\right) + O(\varepsilon) \end{aligned} \tag{24}$$

Simplifying the middle (...) S_{21} term using (18d):

$$\left(A_{12}A_{22}^{-1}\Pi_{z2}^{(0)} - A_{12}\Pi_{z2}^{(0)} S_{22}^{-1} - B_1 L_{g2} S_{22}^{-1}\right) S_{21} \tag{25}$$
$$= \left(A_{12}A_{22}^{-1}\left(\Pi_{z2}^{(0)} S_{22} - A_{22}\Pi_{z2}^{(0)}\right) S_{22} - B_1 L_{g2} S_{22}^{-1}\right) S_{21} \tag{26}$$
$$= \left(A_{12}A_{22}^{-1} B_2 L_{g2} S_{22}^{-1} - B_1 L_{g2} S_{22}^{-1}\right) S_{21} \tag{27}$$
$$= -\left(B_1 - A_{12}A_{22}^{-1}B_2\right) L_{g2} S_{22}^{-1} S_{21} \tag{28}$$

Substituting (27) back into (23) and factorising we have:

$$\left(A_{11} - A_{12}A_{22}^{-1}A_{21}\right)\Pi_{x1}^{(0)} + (B_1 - A_{12}A_{22}^{-1}B_2)(L_{g1} - L_{g2}S_{22}^{-1}S_{21}) \tag{29}$$
$$= \Pi_{x1}^{(0)}\left(S_{11} - S_{12}S_{22}^{-1}S_{21}\right) + O(\varepsilon) \tag{30}$$

which is of $O(\varepsilon)$ error from (15). Thus as $\varepsilon \to 0$, (29) approaches (15), and the matrix $\Pi_{x1} \to \Pi_{x1}^{(0)} = \tilde{\Pi}$.

3 Next Steps

- Expression for Π_{x2} (Is it needed?)
- Higher order epsilon expansion?
- Assumptions, and rigour?

4 S_{11} and S_{22} are 0

$$A_{11}\Pi_{x1} + A_{12}\Pi_{z1} + B_1L_{g1} = \frac{1}{\varepsilon}\Pi_{x2}S_{21}, \tag{31a}$$
$$A_{11}\Pi_{x2} + A_{12}\Pi_{z2} + B_1L_{g2} = \Pi_{x1}S_{12}, \tag{31b}$$
$$\frac{1}{\varepsilon}(A_{21}\Pi_{x1} + A_{22}\Pi_{z1}) + \frac{1}{\varepsilon}B_2L_{g1} = \frac{1}{\varepsilon}\Pi_{z2}S_{21}, \tag{31c}$$
$$\frac{1}{\varepsilon}(A_{21}\Pi_{x2} + A_{22}\Pi_{z2}) + \frac{1}{\varepsilon}B_2L_{g2} = \Pi_{z1}S_{12}. \tag{31d}$$

Similar approach to before, we'll have $\Pi_{x2}^{(0)} = 0$.
Considering zeroth order terms:

$$A_{11}\Pi_{x1}^{(0)} + A_{12}\Pi_{z1}^{(0)} + B_1L_{g1} = \Pi_{x2}^{(1)}S_{21}, \tag{32a}$$
$$A_{12}\Pi_{z2}^{(0)} + B_1L_{g2} = \Pi_{x1}^{(0)}S_{12}, \tag{32b}$$
$$A_{21}\Pi_{x1}^{(0)} + A_{22}\Pi_{z1}^{(0)} + B_2L_{g1} = \Pi_{z2}^{(0)}S_{21}, \tag{32c}$$
$$A_{22}\Pi_{z2}^{(0)} + B_2L_{g2} = 0 \tag{32d}$$

Assuming A_{22} is invertible:

$$\Pi_{z2}^{(0)} = -A_{22}^{-1}B_2L_{g2} \tag{33}$$

Assuming S_{12} is invertible, (33) $\to$ (32b):

$$\Pi_{x1}^{(0)} = (-A_{12}A_{22}^{-1}B_2L_{g2} + B_1L_{g1})S_{12}^{-1} \tag{34}$$

And $\Pi_{z1}^{(0)}$ can be extracted from (32c). Simulation results doesnt seem to agree. Might be implementation issue.

5 Thoughts

- Canonical form - is there a coordinate change that we can exploit?
- Making a family of approximations? what to do when A_{22} is not Hurwitz?
- Realisation?
- Controllability and observability conditions